Movie Recommendation System

Abstract

This project implements a simple Movie Recommendation System using item-based collaborative filtering. A small sample dataset of movies and user ratings is used to demonstrate building an item-user matrix, computing item similarity (cosine), and generating top-N recommendations for users.

Objectives

1. Build a baseline recommendation system that suggests movies to users based on historical ratings. 2. Demonstrate item-based collaborative filtering and similarity computation. 3. Produce reproducible code and documentation for GitHub/Google Drive upload.

Methodology

1. Data: `movies.csv` (movieId, title, genres) and `ratings.csv` (userId, movieId, rating, timestamp). 2. Create an item-user rating matrix where rows are movies and columns are users, filling missing ratings with zeros. 3. Compute item-to-item similarity using cosine similarity on the item-user matrix. 4. For a given user, aggregate scores for candidate movies as the weighted sum of similarities times the user's ratings.

Implementation

The implementation uses Python with `pandas`, `scikit-learn`, and `scipy`. `movie_recommender.py` provides functions to load data, build matrices, compute similarities, and generate recommendations. The script prints example recommendations for users present in the dataset.

Results (Sample)

userId	Top-3 Recommendations
1	Pulp Fiction (1994), Forrest Gump (1994), Interstellar (2014)
2	The Godfather (1972), The Dark Knight (2008), Fight Club (1999)
3	The Matrix (1999), The Godfather (1972), Interstellar (2014)
4	The Dark Knight (2008), The Shawshank Redemption (1994), The Godfather (1972)
5	The Godfather (1972), Pulp Fiction (1994), Forrest Gump (1994)
6	The Godfather (1972), The Shawshank Redemption (1994), Inception (2010)

Conclusion

This project demonstrates a baseline item-based collaborative filtering recommender. For production systems, larger datasets (e.g., MovieLens), normalization, handling sparsity, implicit feedback, and evaluation metrics (RMSE, precision@k, recall@k) should be added. The code and report are provided for GitHub or Google Drive upload.

Files included

- movies.csv
- ratings.csv
- movie_recommender.py
- README.md